

A
Seminar Report
On
**PISNet: A Patient-Independent
1D-CNN Framework for Early
Seizure Prediction Using Transfer
Learning**

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CERTIFICATE

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Abstract

Epilepsy is a chronic neurological condition marked by recurrent and unpredictable seizures, often leading to significant disruptions in patients' health and daily life. Accurate and early prediction of seizures can enhance patient safety and enable timely medical intervention. This work proposes PISNet, a deep learning-based framework for early seizure prediction using scalp electroencephalogram (EEG) signals. The system employs a one-dimensional Convolutional Neural Network (1D-CNN) trained on standardized EEG segments to classify brain activity into interictal and preictal states.

Two evaluation strategies were investigated: a patient-independent generalized model and a patient-dependent Leave-One-Patient-Out (LOPO) approach. The generalized model assesses the network's ability to generalize across subjects, while the LOPO strategy mimics real-world clinical conditions by excluding one patient from training and using their data solely for testing. To improve performance, fine-tuning was applied using a limited number of EEG segments (ranging from 100 to 2000) from the test subject.

Experiments were conducted on the publicly available CHB-MIT Scalp EEG Database, using 18 selected channels downsampled to 64 Hz. The proposed model achieved a test accuracy of 98.5%, sensitivity of 98.9%, and an Area Under the ROC Curve (AUC) of 0.998. Compared to existing state-of-the-art approaches, PISNet achieves 5-7% higher accuracy, establishing its superiority in generalized seizure prediction. These results indicate the framework's strong potential for real-time, patient-specific seizure prediction and provide a promising foundation for future implementation on hardware platforms through VLSI design.

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List of Abbreviations

CNN	Convolutional Neural Network
EEG	Electroencephalogram
1D-CNN	One-Dimensional Convolutional Neural Network
CHB-MIT	Children's Hospital Boston - Massachusetts Institute of Technology
RTL	Register Transfer Level
FPGA	Field Programmable Gate Array
HDL	Hardware Description Language
ASIC	Application-Specific Integrated Circuit

Chapter 1

Introduction

1.1 Background

Epilepsy is a chronic, non-communicable neurological disorder affecting nearly 50 million people worldwide, making it one of the most prevalent neurological conditions [1]. It is characterized by recurrent, unprovoked seizures—brief episodes of excessive and abnormal neuronal activity in the brain. These seizures may manifest as involuntary muscle movements, sensory disturbances, altered awareness, or even loss of consciousness.

The clinical manifestation of epileptic seizures exhibits significant inter-individual variability in both temporal dynamics and symptom severity. While some patients present with mild, infrequent episodes, others experience severe and recurrent seizures that profoundly impact neurological function and quality of life. This unpredictability necessitates advanced prediction systems capable of accounting for: (1) heterogeneous seizure semiologies, (2) variable preictal durations (30–90 minutes), and (3) patient-specific electrophysiological signatures. Contemporary deep learning approaches [2] have demonstrated particular promise in addressing these challenges through automated feature learning from multi-channel EEG recordings, achieving superior performance compared to traditional spectral analysis methods [3].

1.1.1 Proposed Pipeline Overview

The overall pipeline developed in this project for seizure detection using EEG signals is illustrated in Figure 1.1. The system begins with pre-processing EEG recordings from the CHB-MIT dataset, where raw signals are segmented and normalized. A lightweight 1D Convolutional Neural Network (1D-CNN) is then trained to classify whether a given segment corresponds to an interictal or preictal state.

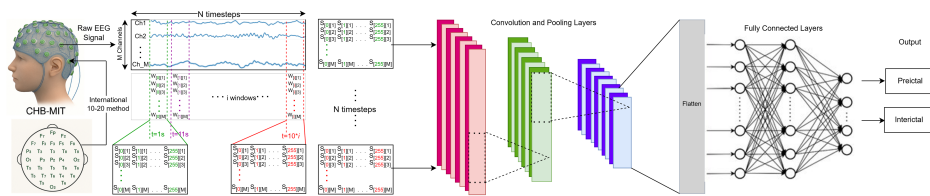


Figure 1.1: General pipeline of the seizure prediction system developed in this project.

1.2 EEG Signal Fundamentals

EEG (Electroencephalography) is a technique that captures the electrical activity of the brain through electrodes placed on the scalp. It provides a non-invasive window into neural dynamics by recording the summed postsynaptic potentials of thousands of neurons. These electrical signals vary in frequency and amplitude, with different patterns associated with specific physiological and cognitive states.

EEG signals are commonly divided into five major frequency bands: Delta (δ), Theta (θ), Alpha (α), Beta (β), and Gamma (γ). Each brain wave is linked to particular functional states:

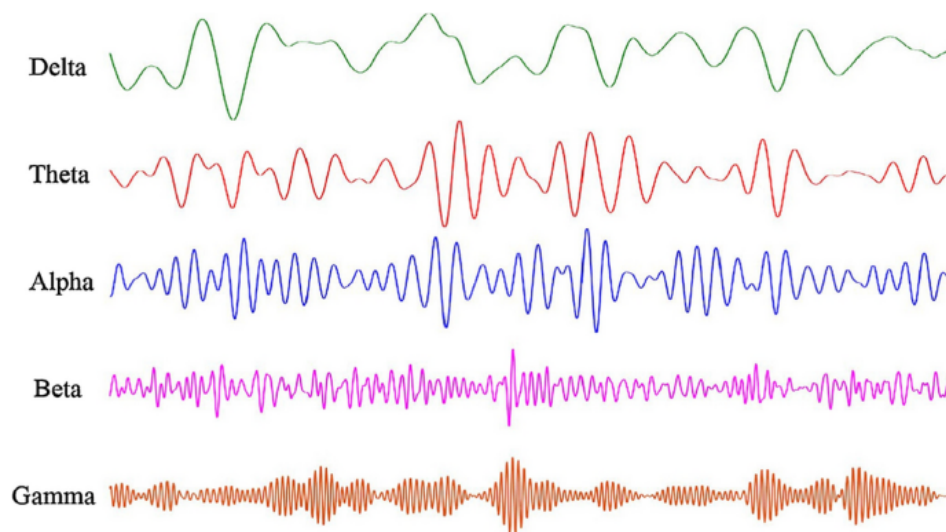


Figure 1.2: EEG frequency bands and their associated brain states

EEG signals are characterized by five principal frequency bands, each associated with distinct cognitive and physiological states (Table 1.1):

1.2.1 EEG Acquisition Methods

Electroencephalography (EEG) is the primary tool for monitoring brain activity associated with seizures. The word Electroencephalogram (EEG) originates from Greek:

- “Electro” – relating to electrical activity
- “Encephalo” – meaning brain
- “Gram” – meaning written record

There are several EEG acquisition methods:

In this project, we utilize the CHB-MIT Scalp EEG Database, which includes:

Table 1.1: Characteristics of EEG Frequency Bands

Wave	Frequency	Amplitude	Associated States
Delta (δ)	0.5–4 Hz	20–200 μ V	Deep sleep, unconscious states
Theta (θ)	4–8 Hz	10–50 μ V	Drowsiness, meditation, light sleep
Alpha (α)	8–13 Hz	20–60 μ V	Relaxed wakefulness (eyes closed)
Beta (β)	13–30 Hz	5–20 μ V	Active concentration, alertness
Gamma (γ)	> 30 Hz	2–10 μ V	Sensory integration, cognition

Table 1.2: EEG Acquisition Methods

Type	Description
Scalp EEG	Non-invasive method using the international 10-20 system. Most common due to safety and ease of use.
Intracranial EEG (iEEG)	Invasive technique with electrodes placed directly on brain tissue. Higher spatial resolution.
Video-EEG Monitoring	Combines EEG with synchronized video for correlation of electrical signals with physical symptoms.

- Scalp EEG signals from pediatric patients
- 23 electrodes (10-20 system)
- Long-term, multi-channel recordings
- Detailed seizure annotations

1.2.2 Understanding Seizure States

Seizures are sudden, uncontrolled electrical disturbances in the brain that can lead to changes in behavior, movement, sensations, and consciousness. Electroencephalography (EEG) is commonly used to monitor brain activity and detect abnormal patterns associated with epilepsy. For seizure prediction, it is essential to understand the various brain states that occur before, during, and after a seizure, as shown below in Figure 1.3.

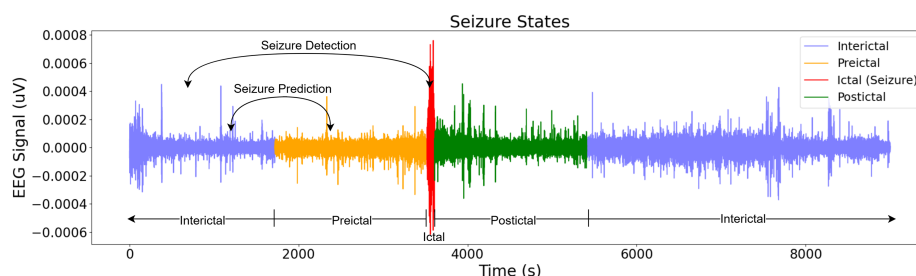


Figure 1.3: Transition through seizure states (Interictal → Preictal → Ictal → Postictal) with corresponding EEG patterns and clinical relevance for prediction vs detection.

1. Interictal State:

The interictal state refers to the period between seizures when the brain exhibits relatively normal electrical activity. This state is generally long and neurologically stable. EEG segments from this phase are typically used as representations of the "non-seizure" class in classification tasks.

2. Preictal State:

The preictal state is the critical window leading up to a seizure, typically occurring 30 to 90 minutes before onset. Subtle yet significant changes in EEG patterns often arise in this phase, making it essential for prediction. In this study, a 30-minute preictal window is defined and used to label EEG segments preceding seizure onset.

3. Ictal State:

The ictal state represents the seizure event itself. It is marked by a sudden burst of abnormal, high-amplitude, and high-frequency brain activity. While traditional detection models focus on this phase, our goal is to anticipate the seizure—thus, emphasis is placed on analyzing the preictal state.

4. Postictal State:

The postictal state follows the seizure and can last from minutes to hours. It is characterized by brain recovery, fatigue, and disorientation. Although this state is not directly used in training, it provides insight into the overall seizure timeline and recovery.

Among these phases, the **preictal state** is especially important for seizure prediction, as it provides a time window where early intervention can significantly reduce risk. Accurate forecasting enables patients to take preventive steps such as moving to a safe environment, taking medication, or notifying caregivers—thereby enhancing safety and peace of mind.

1.2.3 Existing Approaches to Seizure Analysis

Seizure analysis using EEG data can be broadly categorized into three primary approaches:

1. **Seizure Prediction:**

This approach involves identifying the *preictal state* before the actual seizure onset. It holds the highest clinical value as it allows for timely and preventive action to be taken. However, it is also the most challenging, as the EEG patterns of the preictal and interictal states are often very similar, making accurate classification difficult (as illustrated in Figure 1.3).

2. **Seizure Detection:**

This focuses on recognizing ongoing seizure activity, also known as the *ictal state*. Detection systems are commonly used in clinical settings to identify and log seizures but do not provide any advance warning, which limits their utility for real-time safety and intervention.

3. **Seizure Classification:**

This method deals with differentiating types or origins of seizures, such as focal versus generalized seizures. It is mainly used for diagnostic purposes and to guide treatment strategies in long-term monitoring.

Among these, **seizure prediction** remains the most impactful yet technically demanding approach due to the highly variable and subtle nature of preictal EEG features across patients.

1.3 Motivation

The motivation for this project lies in both the critical healthcare need and the opportunity to apply cutting-edge deep learning methods in biomedical signal analysis. The sudden and unpredictable onset of seizures affects the daily lives of millions, often resulting in injury, anxiety, and loss of autonomy.

Existing systems predominantly focus on *detection* rather than *prediction*, offering no scope for preventive action. Additionally, many solutions are heavily reliant on patient-specific training, which limits their generalizability and scalability in real-world applications.

Deep learning offers a promising alternative. It enables automatic learning from raw EEG data and can extract intricate temporal and spatial features without the need for manual preprocessing. These capabilities support the development of both generalized (cross-patient) and personalized models, effectively addressing many limitations of conventional approaches and paving the way for scalable seizure prediction solutions.

1.4 Objective

The primary objective of this project is to design and evaluate a deep learning-based framework for early seizure prediction using scalp EEG data. Specifically, the aim is to develop a 1D Convolutional Neural Network (1D-CNN) model capable of distinguishing between interictal and preictal brain states, with a prediction horizon of 30 minutes prior to seizure onset.

The key objectives include:

1. Developing a generalized seizure prediction model that works across patients using the CHB-MIT Scalp EEG dataset.
2. Incorporating transfer learning to fine-tune the generalized model using limited patient-specific data for enhanced personalization.
3. Minimizing false alarms while maintaining high sensitivity and accuracy in seizure prediction.

1.5 Scope of the Work

The scope of this project encompasses the offline development, training, and evaluation of a seizure prediction system using EEG signals. The major tasks covered are:

- Preprocessing EEG signals and segmenting them into interictal and preictal windows.
- Designing a 1D-CNN architecture to learn temporal patterns in EEG sequences.
- Implementing a patient-independent training scheme and applying fine-tuning for patient-specific enhancement.
- Evaluating model performance on the CHB-MIT dataset using accuracy, sensitivity, and AUC metrics.
- Ensuring the model's scalability and applicability for future real-time deployment on hardware such as VLSI platforms.

1.6 Contribution

The major contributions of this work are summarized below:

1. We propose a patient-independent seizure prediction model (PISNet) based on 1D-CNN that achieves high performance in detecting preictal activity up to 30 minutes in advance.
2. We explore the use of transfer learning to personalize the generalized model with a small amount of target patient data, improving prediction accuracy without re-training from scratch.
3. Our method eliminates the need for handcrafted features by learning directly from raw EEG signals, offering a fully automated prediction pipeline.
4. The work sets the foundation for future deployment of seizure prediction systems on portable and low-power platforms for real-time health monitoring.

Chapter 2

Literature Review

Epileptic seizure prediction has become a vital research area within biomedical signal processing, especially with the rising integration of deep learning in healthcare. Accurate and early seizure prediction can significantly enhance the quality of life for epilepsy patients by providing timely warnings and enabling preventive interventions. In recent years, researchers have extensively explored both classical machine learning (ML) and deep learning (DL) approaches for seizure prediction using electroencephalogram (EEG) data. Broadly, these studies fall into two major categories: *patient-independent* models and *patient-specific* models.

Patient-specific approaches often yield high performance as they are trained on the unique EEG patterns of individuals. However, they lack scalability for clinical deployment. On the other hand, patient-independent models aim for generalization across multiple subjects, making them more practical for real-world implementation. The major challenge in building such models lies in the high variability of EEG patterns across individuals due to physiological, anatomical, and recording differences.

2.1 Patient-Independent Seizure Prediction

Patient-independent models are designed to predict seizures across multiple subjects using a single trained model. These approaches are more scalable and suitable for deployment in clinical settings where individualized training may not be feasible. However, such models face significant challenges in generalizing across the high inter-subject variability in EEG data.

Tsiouris et al. [4] proposed a patient-independent seizure prediction framework using a Support Vector Machine (SVM). The SVM was trained on a composite feature set that included time-domain measures (e.g., mean, variance), frequency-domain metrics (e.g., spectral entropy), and graph-theoretical properties derived from EEG connectivity. Despite this comprehensive feature set, their system achieved an accuracy of only 68.50% on the CHB-MIT dataset, indicating the limitations of traditional machine learning techniques in capturing generalizable EEG patterns.

Khan et al. [5] addressed this issue using deep learning by proposing a Convolutional Neural Network (CNN)-based model. Their system achieved a Receiver Operating Characteristic Area Under the Curve (ROC-AUC) of 0.8660 for detecting preictal states within a 10-minute horizon. This study demonstrated that deep learning architectures could learn abstract representations directly from raw EEG, eliminating the need for hand-crafted features. Nevertheless, the performance was still inconsistent across

different subjects, emphasizing the inherent difficulty in achieving robust generalization.

Roy et al. [6], in their comprehensive survey, observed that while patient-specific systems typically perform better due to their tailored nature, the field is gradually moving toward patient-independent solutions. They noted that these models, though harder to train and validate, are essential for practical deployment in ambulatory EEG monitoring systems.

2.2 Patient-Specific Seizure Prediction

In contrast, patient-specific models are tailored for individual subjects by training on their own historical EEG recordings. These models leverage personalized patterns in the brain's electrical activity to distinguish between interictal (normal) and preictal (seizure-impending) states.

Daoud and Bayoumi [7] proposed a hybrid architecture combining Convolutional Neural Networks (CNNs) and Bidirectional Long Short-Term Memory (BiLSTM) networks. This model was designed to capture both spatial and temporal EEG features, achieving nearly 100% accuracy across selected subjects in the CHB-MIT dataset. They further introduced a channel selection mechanism to optimize model efficiency and reduce computational complexity.

Zhang et al. [8] tackled the problem of limited training data per patient by employing a novel data augmentation strategy. They used a Generative Adversarial Network (GAN) to synthesize EEG segments, combined with multi-segment slicing to enhance temporal resolution. Their model achieved an accuracy of 92.2% with a prediction window of 30 minutes. However, the use of GANs introduced additional training overhead and complexity, which could hinder deployment in real-time or resource-constrained settings.

Tsiouris et al. [4], in another work, used Long Short-Term Memory (LSTM) networks to develop a patient-specific model trained on time-domain statistics, spectral power, and autocorrelation features. They reported an exceptional performance of 99.84% accuracy, highlighting the effectiveness of deep recurrent models in capturing the temporal dynamics of EEG signals.

2.3 Challenges and Gaps in Existing Work

Despite the success of patient-specific models, their applicability is limited due to the need for subject-wise training and calibration. This makes them less viable for broader

clinical deployment, especially when dealing with new patients or large-scale monitoring systems.

On the other hand, while patient-independent models offer a path toward scalable and universal seizure prediction systems, they often struggle to maintain performance across diverse subjects. Roy et al. [6] and Rasheed et al. [9] emphasized that deep learning models, although accurate, often act as black boxes with limited interpretability. This lack of transparency makes it difficult for clinicians to trust and adopt these systems.

Furthermore, Kuhlmann et al. [10] pointed out that many existing studies fail to systematically evaluate how specific features influence model decisions. There is a growing need for explainable AI models that not only predict accurately but also provide insights into the underlying neural dynamics associated with seizures.

2.4 Our Contribution in Context

To bridge these gaps, our study presents a novel patient-independent seizure prediction framework using a 1D Convolutional Neural Network (1D-CNN) trained on raw EEG data from the CHB-MIT Scalp EEG dataset. The key aspects of our approach include:

- **Generalization without Hand-Crafted Features:** Unlike traditional methods that rely on manually extracted features or complex preprocessing, our model learns directly from the raw EEG signal, reducing preprocessing overhead and allowing end-to-end learning.
- **Minimal Personalization via Transfer Learning:** We incorporate a lightweight transfer learning mechanism that fine-tunes the model using a small subset of patient-specific data. This provides a balance between generalization and personalization, enabling improved adaptability without requiring full retraining.
- **Hardware-Efficient Implementation:** By limiting the model to 18 key EEG channels and using downsampled data, we ensure that the training and inference processes are computationally efficient. This paves the way for potential integration into wearable or edge-computing devices for continuous monitoring.

In summary, our work contributes to the field by proposing a scalable, interpretable, and practically deployable seizure prediction system. It directly addresses the core limitations in prior studies—namely, lack of generalizability, over-reliance on feature engineering, and inefficiency in real-time applications—thereby making a meaningful step toward real-world clinical adoption.

Chapter 3

Methodology

3.1 Introduction to Methodology

The primary aim of this research is to design and implement a robust, accurate, and computationally efficient pipeline for the early prediction of epileptic seizures using electroencephalogram (EEG) signals. Epileptic seizures are sudden and often unpredictable, posing significant health risks and impacting the quality of life for patients. Accurate prediction can enable timely intervention, offering patients a better chance to prevent injury and take precautionary measures.

This methodology chapter outlines the systematic approach adopted in this study to build and evaluate a deep learning-based seizure prediction framework. It is structured into two key stages:

1. **Preprocessing of EEG signals** — where raw EEG recordings are cleaned, segmented, normalized, and labeled into interictal and preictal states.
2. **Model design and training** — where a one-dimensional Convolutional Neural Network (1D-CNN) is developed, trained, and validated using both patient-independent and patient-dependent strategies.

Two modeling paradigms are explored:

- **Patient-Independent Modeling:** The model is trained across multiple patients and tested on unseen individuals to assess its generalizability. This approach is critical for real-world deployment, where patient-specific data may not be available.
- **Patient-Dependent Modeling:** The model is trained and evaluated for each individual patient. This personalized approach often achieves higher prediction accuracy and serves as a benchmark to gauge improvements from generalization.

The methodology involves several steps: raw data acquisition, signal cleaning, feature preparation, model architecture design, training pipeline setup, and evaluation. Performance is assessed using standard classification metrics — **Accuracy**, **Sensitivity (Recall)**, and **Specificity** — to provide a comprehensive view of the system’s effectiveness.

Finally, the pipeline is designed to be lightweight and modular, making it compatible for eventual hardware acceleration and real-time deployment on edge devices such as FPGAs.

3.2 Dataset Description

Dataset Source

The dataset used in this study is the **CHB-MIT Scalp EEG Database**, a publicly available collection of pediatric EEG recordings hosted on PhysioNet. It is one of the most widely used datasets in seizure detection and prediction research.

The data were collected at the Children’s Hospital Boston (CHB) as part of long-term monitoring for epilepsy surgery evaluation. Patients with drug-resistant epilepsy were continuously monitored, and medications were often withdrawn to allow spontaneous seizures to occur for observation and diagnosis.

Summary of Dataset Characteristics

Table 3.1: CHB-MIT Scalp EEG Dataset Summary

Type	Description
Subjects	22 pediatric patients (5 males aged 3–22 and 17 females aged 1.5–19); 1 subject recorded twice (total 23 cases)
Files	664 EEG recordings in EDF format; Each case has 9–42 files, most of which are 1 hour long
Long-duration files	Some exceptions (e.g., chb10: 2-hour, chb04/chb09/chb23: 4-hour recordings)
Sampling Rate	256 Hz with 16-bit resolution
EEG Channels	23–26 channels per file using the International 10-20 Electrode Placement System
Annotations	198 seizures annotated with start and end times (in seconds)
Additional Signals	ECG and VNS markers included in a few files
Noise	Dummy signals (denoted by “-”) included in some files, ignored in analysis

Relevance to Our Study

This dataset is particularly suitable for seizure prediction due to the following:

- High temporal resolution and long-term continuous recordings
- Detailed expert-labeled seizure annotations

- Variability in patient demographics and seizure types
- Availability of interictal and preictal windows spanning multiple hours

For our project, we selected 12 representative patients based on annotation completeness, seizure frequency, and channel consistency. Only EEG channels were used; ECG or auxiliary channels were discarded.

3.3 EEG Preprocessing Pipeline

Overview

Raw EEG signals contain various types of noise, inconsistencies, and irrelevant components. A well-structured preprocessing pipeline is essential to extract clean, discriminative features from the signals. Our preprocessing steps are illustrated in Figure 3.1.

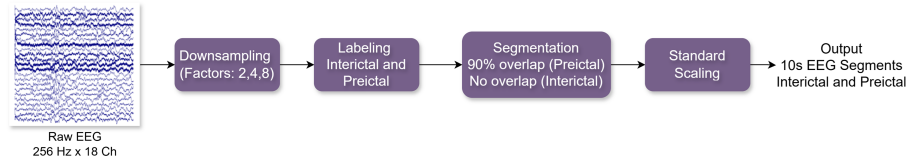


Figure 3.1: Overview of the EEG preprocessing pipeline

3.3.1 Raw EEG Acquisition

The original EEG recordings are sampled at 256 Hz across 23–26 channels per patient. For this study, we consistently use 18 bipolar channel pairs that appear across all subjects, grouped by brain regions:

Table 3.2: Selected EEG Channel Pairs for Analysis

Region	Channel Pairs
Left Hemisphere	FP1-F7, F7-T7, T7-P7, P7-O1
Frontal-Left	FP1-F3, F3-C3, C3-P3, P3-O1
Frontal-Right	FP2-F4, F4-C4, C4-P4, P4-O2
Right Hemisphere	FP2-F8, F8-T8, T8-P8, P8-O2
Midline	FZ-CZ, CZ-PZ

3.3.2 Downsampling

To reduce the computational load while retaining essential features, all signals are downsampled from 256 Hz to 64 Hz. This process is performed using anti-aliasing filters to prevent distortion and preserve frequency characteristics important for seizure onset patterns.

3.3.3 Labeling of Segments

Each EEG file is segmented into labeled windows based on seizure proximity:

- **Preictal:** Segments within 30 minutes prior to seizure onset.
- **Interictal:** Segments at least 4 hours before or 30 minutes after any seizure. Postictal windows are excluded to avoid confounding patterns.

3.3.4 Segmentation

The preprocessed EEG signal is divided into 10-second segments:

- **Preictal Segments:** Extracted with 90% overlap to increase sample count and maintain high temporal resolution.
- **Interictal Segments:** Extracted with no overlap to ensure a diverse, balanced dataset without redundancy.

Each segment corresponds to a tensor of shape $(640, 18)$ — 640 time points per channel.

3.3.5 Normalization

To eliminate inter-channel amplitude bias and enhance convergence during training, we apply Z-score normalization:

$$Z = \frac{X - \mu}{\sigma}$$

Where:

- X = raw EEG signal
- μ = mean of each channel
- σ = standard deviation of each channel

All segments are normalized channel-wise to have zero mean and unit variance.

3.4 Model Architecture and Training Pipeline

Overview

We employ a 1D Convolutional Neural Network (CNN) tailored to extract temporal patterns from EEG time-series data. The CNN architecture is lightweight, modular, and suitable for embedded implementations. The model receives a 10-second EEG segment and outputs the probability of it belonging to the preictal or interictal class.

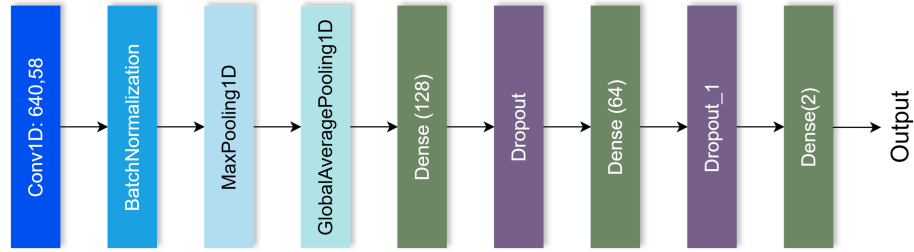


Figure 3.2: Block diagram of the model training and classification pipeline

3.4.1 Input Design

Each input segment has the shape $(640, 18)$ — representing 10 seconds of data at 64 Hz across 18 EEG channels. The input is processed as a multivariate time-series.

3.4.2 CNN Architecture

The model consists of the following components:

- **1D Convolution Layer:** 58 filters, kernel size 3, ReLU activation, and ‘same’ padding
- **Batch Normalization:** Ensures stable and faster training
- **MaxPooling1D:** Pool size 2 to reduce temporal dimensionality
- **Global Average Pooling:** Summarizes feature maps into one vector, reducing parameters
- **Dense Layer 1:** 128 units with ReLU and Dropout(0.5)
- **Dense Layer 2:** 64 units with ReLU and Dropout(0.5)
- **Output Layer:** 2 units with Softmax for binary classification

3.4.3 Training Strategy

We use two training approaches:

1. Generalized (Cross-Subject) Modeling

- A portion (90%) of data from all subjects is used for training; 10% is used for testing.
- This tests the model's ability to generalize across different patients.

2. Leave-One-Patient-Out (LOPO) Modeling

- In each iteration, data from one subject is held out as the test set.
- The remaining data is split 80:20 for training and validation.
- The base model is then fine-tuned using 100, 500, 1000, and 2000 labeled samples from the test subject to simulate transfer learning.

Chapter 4

Results and Discussion

This chapter presents the evaluation of the proposed seizure prediction system under two distinct modeling strategies: a **Patient-Independent Generalized Model** and a **Patient-Specific Leave-One-Patient-Out (LOPO) Model**. Each approach was designed to assess different aspects of model performance — the former focuses on the model’s ability to generalize across patients, while the latter simulates personalized seizure prediction using limited patient-specific data.

4.1 Patient-Independent Generalized Model (Cross-Subject Evaluation)

In the generalized setup, EEG data from all subjects was aggregated. For each subject, 10% of the data was randomly selected for testing, and the remaining 90% was used for training. This configuration allowed the model to learn seizure-related patterns across a diverse population and assess its generalization capability on unseen samples from the same overall distribution.

This approach is critical for applications where subject-specific training data is not available, such as wearable devices used by new patients. While accuracy might be slightly lower due to inter-subject variability, it provides insight into the robustness and transferability of the trained model.

4.2 Patient-Specific Leave-One-Patient-Out (LOPO) Model

To emulate a real-world clinical scenario with limited individual data, a LOPO cross-validation strategy was employed. In each iteration, one subject was entirely excluded from training and used solely for testing. The model was trained using an 80-20 training-validation split on the remaining subjects.

To enable personalization and simulate transfer learning, a fine-tuning phase was introduced. In this step, EEG segments from the excluded subject were incrementally added to the training set in the amount of 100 samples, 500 samples, 1000 samples and 2000 samples.

This allowed the model to gradually adapt to the subject’s unique EEG characteristics. Performance was evaluated at each stage to assess the effect of personalization on prediction accuracy and robustness. Results demonstrated that even a small amount of

subject-specific data significantly improved classification accuracy, making the model suitable for adaptive deployment in clinical settings.

To assess the performance of the proposed seizure prediction model, several evaluation metrics are used. These metrics provide a comprehensive understanding of the model's behavior in correctly identifying seizure-prone (preictal) and normal (interictal) brain states.

4.2.1 Segment-Based Performance Metrics

Confusion Matrix Definitions The following notations are used:

- **TP (True Positive):** Number of preictal segments correctly classified as preictal.
- **TN (True Negative):** Number of interictal segments correctly classified as interictal.
- **FP (False Positive):** Number of interictal segments incorrectly classified as preictal.
- **FN (False Negative):** Number of preictal segments incorrectly classified as interictal.

Performance Metrics

- **Accuracy (ACC):**

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Measures the overall correctness of the model.

- **Sensitivity (Recall or True Positive Rate - TPR):**

$$\text{Sensitivity} = \frac{TP}{TP + FN}$$

Indicates the model's ability to correctly identify preictal segments.

- **Specificity (True Negative Rate - TNR):**

$$\text{Specificity} = \frac{TN}{TN + FP}$$

Measures the model's capability to avoid false alarms.

4.2.2 Event-Based Performance Metrics

- **Event-Based Sensitivity:**

$$\text{Sensitivity}_{\text{event}} = \frac{\text{Number of correctly predicted seizures}}{\text{Total number of seizures}}$$

Measures the model’s ability to predict entire seizure events (not just segments).

- **Event-Based False Positive Rate:**

$$\text{FPR}_{\text{event}} = \frac{\text{Number of false predictions}}{\text{Hours of interictal iEEG}}$$

Quantifies false alarms per hour of normal brain activity.

These metrics ensure a detailed evaluation of the model’s performance at both segment and event levels. Particular attention is given to minimizing false positives (to avoid unnecessary alarms) and maximizing sensitivity (to ensure timely seizure prediction), while the event-based metrics provide clinical relevance by evaluating whole seizure prediction performance.

4.3 Patient-Independent Generalized Model

In the generalized model setup, 10% of each subject’s data is set aside for testing while the remaining 90% is used for training. This strategy helps evaluate how well the model generalizes across multiple subjects without being fine-tuned to any single individual.

We carried out multiple experiments under this setup by varying model and training parameters to identify optimal configurations. Each variation is discussed in the subsections below.

4.3.1 Effect of Sampling Frequency

In this experiment, we analyze how different sampling frequencies — specifically 128 Hz, 64 Hz, and 32 Hz — affect the performance of the seizure prediction model. The EEG signals were downsampled accordingly, and the model was trained and evaluated under each setting.

Table 4.1 summarizes the results across key performance metrics: training accuracy, validation accuracy, test accuracy, sensitivity, specificity, false positive rate (FPR), AUROC, and F1 Score.

Table 4.1: Effect of EEG Sampling Frequency on Model Performance

Sampling Rate (Hz)	Train Acc.	Val Acc.	Test Acc.	Sens.	Spec.	FPR	AUROC	F1 Score
128 (Down: 2)	91.25%	89.94%	90.26%	85.74%	91.78%	0.0822	0.9600	0.8156
64 (Down: 4)	99.90%	96.37%	96.98%	96.83%	97.03%	0.0297	0.9936	0.9421
32 (Down: 8)	95.16%	90.02%	89.35%	81.20%	92.16%	0.0784	0.9546	0.7962

Observations

- The 64 Hz sampling frequency achieved the best overall performance, with the highest test accuracy (96.98%), sensitivity (96.83%), specificity (97.03%), and F1 score (94.21%). It also yielded the highest AUROC (99.36%), indicating excellent discriminative power between preictal and interictal states.
- While the 128 Hz setup provided a respectable AUROC of 96.00%, its performance was slightly inferior in terms of accuracy and F1 score compared to the 64 Hz setting. The increased resolution may have introduced more noise or redundancy without substantial benefit.
- The 32 Hz configuration showed the lowest overall performance. Although it reduced computational load, it also degraded critical temporal information, resulting in lower sensitivity and F1 score. Nonetheless, its AUROC (95.46%) indicates it still performs significantly better than random classification.

These results suggest that a sampling frequency of 64 Hz offers an optimal trade-off between model accuracy and computational efficiency, making it ideal for real-time and resource-constrained seizure prediction systems.

4.3.2 Effect of Number of Convolutional Layers

This section explores the impact of increasing the number of convolutional layers in the 1D CNN architecture on the model's performance. We trained and evaluated four variants of the model with 1 to 4 convolutional layers, while keeping other parameters (sampling rate = 64 Hz, segment duration = 10 seconds, 18 EEG channels) fixed.

Table 4.2 summarizes the results across key performance metrics: training accuracy, validation accuracy, test accuracy, sensitivity, specificity, false positive rate (FPR), AUROC, and F1 Score.

Observation:

- The architecture with **3 convolutional layers** and filters [32, 64, 128] yielded the best performance, achieving a test accuracy of **98.54%**, sensitivity of **98.51%**, and specificity of **98.56%**, demonstrating excellent generalization and balanced classification.

Table 4.2: Ablation Study: Effect of Convolutional Layer Depth and Filter Sizes on Model Performance

Architecture	Total Params	Trainable Params	Filter Sizes	Test Acc.	Sensitivity	Specificity	FPR
CNN (1 Conv Layer + BN + MaxPool + FC)	14498	14434	[32]	93.66%	95.43%	93.06%	0.0694
CNN (2 Conv Layers + BN + MaxPool + FC)	25058	25866	[32, 64]	96.33%	97.10%	96.07%	0.0393
CNN (3 Conv Layers + BN + MaxPool + FC)	58466	58018	[32, 64, 128]	98.54%	98.51%	98.56%	0.0144
CNN (4 Conv Layers + BN + MaxPool + FC)	174434	173474	[32, 64, 128, 256]	98.31%	97.54%	98.58%	0.0142

- Interestingly, the **2-layer architecture** (filters [32, 64]) also performed competitively with a test accuracy of **96.33%**, only **2.21%** lower than the 3-layer model. This marginal difference suggests that further performance gains may depend more on filter optimization than depth alone.
- The addition of a **fourth convolutional layer** (filters [32, 64, 128, 256]) did not yield significant improvements in test accuracy (**98.31%**) and showed slightly reduced sensitivity, indicating possible overfitting or redundancy.
- The results imply that the transition from 2 to 3 layers provides a critical performance boost, while deeper architectures beyond 3 layers may offer diminishing returns. This motivates a deeper analysis of filter size and feature extraction efficiency rather than simply increasing depth.

4.3.3 Filter Size Tuning

To assess the effect of filter size on classification performance, we conducted an ablation study on the convolutional layer(s) by varying the filter sizes while keeping other model parameters fixed. This evaluation helps in determining the optimal receptive field for seizure-related EEG features.



Figure 4.1: Performance variation with different filter sizes in a two-layer CNN

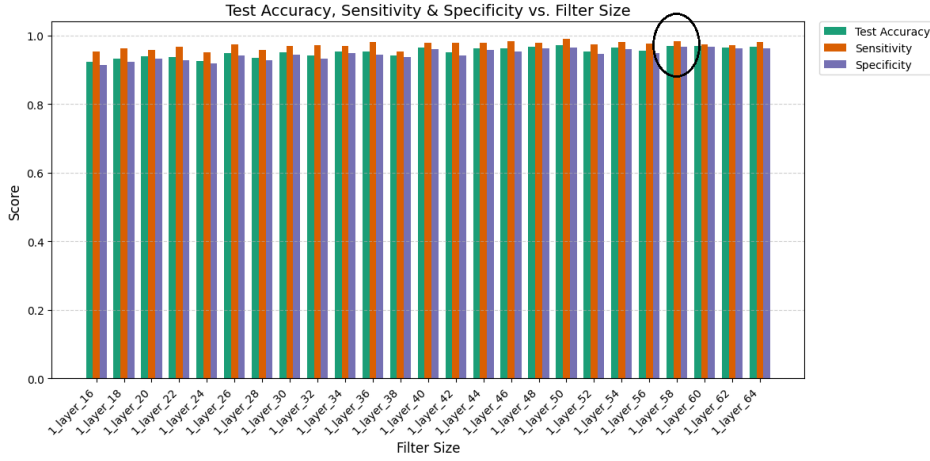


Figure 4.2: Performance variation with different filter sizes in a single-layer CNN

While initial experiments showed that a 2-layer CNN with filters [56, 112] yielded high performance (Test Accuracy: 98.74%, Sensitivity: 98.79%, Specificity: 98.79%), the marginal gain over the 1-layer architecture was not significant enough to justify the added complexity and parameters.

Final Architecture Choice: Based on simplicity, interpretability, and generalization, we selected a **single-layer CNN** for further tuning.

Figure 4.2 illustrates the performance metrics across various filter sizes in the single-layer configuration. Among all candidates, a filter size of **58** achieved the best balance across test accuracy, sensitivity, and specificity.

- **Test Accuracy:** The highest test accuracy of **97.03%** was observed at filter size 58, indicating excellent generalization to unseen EEG data.
- **Sensitivity:** At filter size 58, sensitivity peaked at **97.27%**, demonstrating the model's capability to detect preictal segments reliably.
- **Specificity:** A specificity of **97.16%** was also achieved, reflecting strong performance in minimizing false positives (FPR: 0.0284).

Hence, a **single-layer CNN with filter size 58** is adopted as the final architecture for all further experiments in this study, balancing high predictive performance with computational efficiency.

4.3.4 Hyperparameter Tuning

To optimize the performance of the seizure prediction model, key hyperparameters such as **batch size**, **learning rate**, and **optimizer** were tuned experimentally. The best configuration identified through grid search is summarized in Table 4.3.

Table 4.3: Optimal Hyperparameter Settings

Hyperparameter	Value
Batch Size	32
Learning Rate	0.001
Optimizer	Adam

The resulting model was evaluated using standard classification metrics, as shown in Table 4.4. These results reflect the performance under the optimized configuration.

Table 4.4: Performance Metrics under Optimal Hyperparameters

Metric	Value
Test Accuracy	98.54%
Sensitivity	98.94%
Specificity	98.41%
F1 Score	97.18%
AUC	0.9986
False-Positive Rate	0.0159

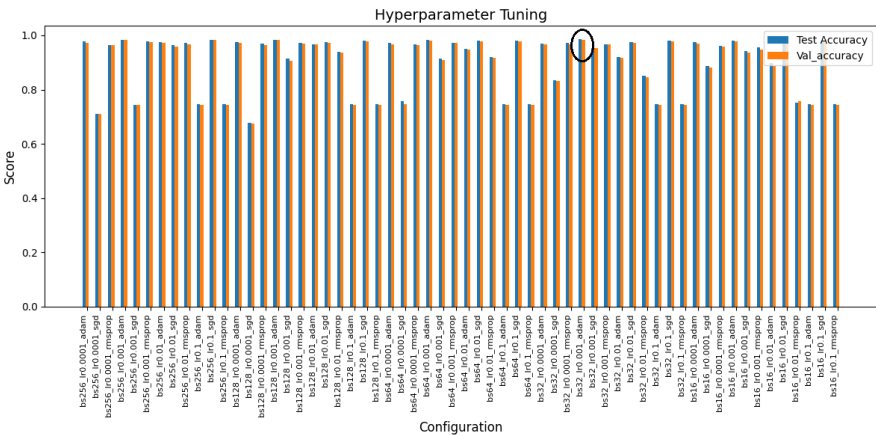


Figure 4.3: Effect of hyperparameter combinations on model performance

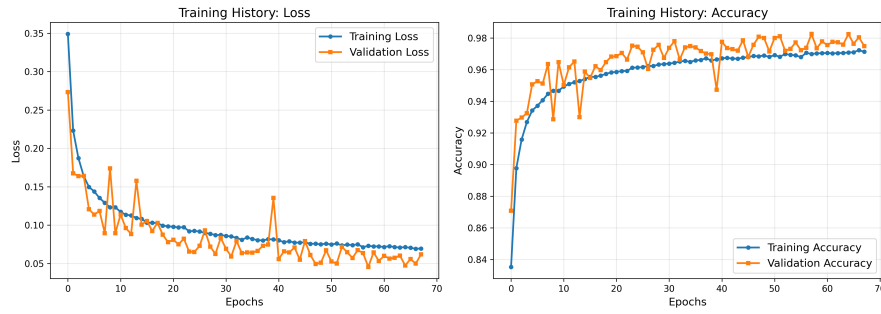


Figure 4.4: Training and validation accuracy/loss over epochs

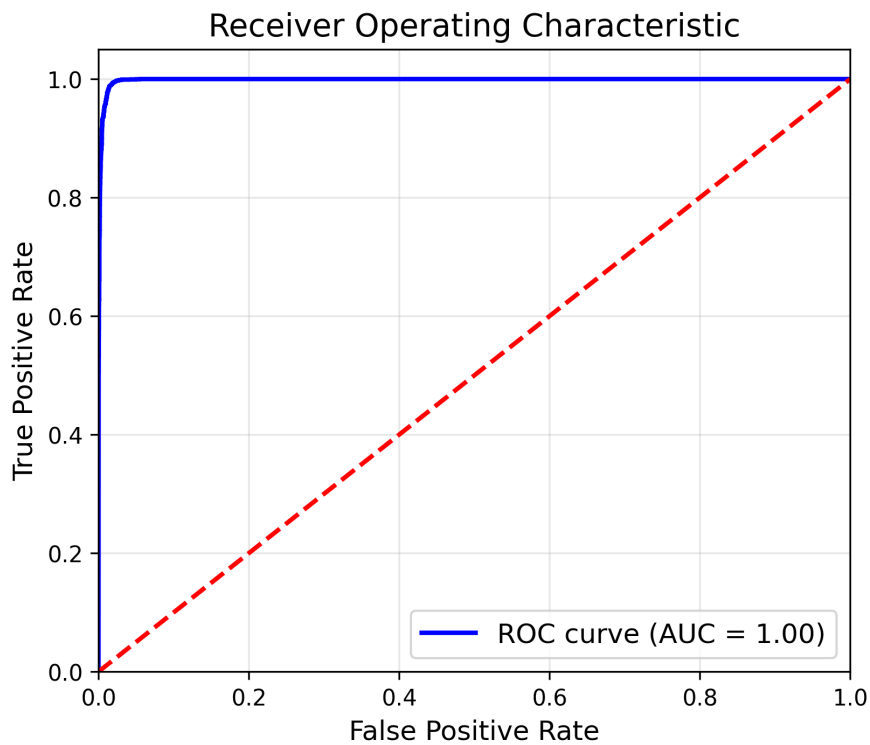


Figure 4.5: Receiver Operating Characteristic (ROC) curve for test set

Observations:

- A batch size of 32 provided a good trade-off between training stability and convergence speed.
- The Adam optimizer consistently outperformed alternatives like SGD and RM-Sprop in terms of both accuracy and F1 score.
- A learning rate of 0.001 led to stable and fast convergence, as illustrated in Figure 4.4.

- The model achieved a test accuracy of **98.54%** with sensitivity of **98.94%** and specificity of **98.41%**, indicating strong and balanced classification performance.
- The AUC of **0.9986** and a low FPR of **0.0159** confirm the model’s excellent discrimination capability and reliability in real-world deployment.

4.4 Patient-Specific Evaluation using Transfer Learning

In real-world clinical settings, patient data is often limited and highly variable. To evaluate the ability of our model to adapt to individual-specific EEG patterns, we implemented a **Patient-Specific Learning** approach using **Transfer Learning (TL)**.

We adopted the **Leave-One-Patient-Out (LOPO)** cross-validation strategy, where one subject is entirely excluded from training and used solely for testing. This simulates a scenario where no prior data from the patient is available during model training. In each iteration:

- The model is trained on data from the remaining 11 subjects.
- The excluded subject’s data is used for evaluation.
- This process is repeated for each of the 12 selected subjects.

To further enhance personalization, we applied **transfer learning** by fine-tuning the trained model using a small number of samples (100, 1000, and 2000) from the excluded subject’s own data. This mimics the real-world scenario where a minimal amount of patient-specific data becomes available post-deployment (e.g., from initial monitoring or clinical tests).

Table 4.5 summarizes the classification accuracy before and after transfer learning for each subject.

Table 4.5: Transfer Learning Results for Selected Subjects

Subject	Total Samples	LOPO Acc. (%)	TL@100 (%)	TL@1000 (%)	TL@2000 (%)	Inf. Time (ms)
chb01	123,283	51.66	91.80	98.40	98.78	0.387
chb03	122,607	79.10	92.42	97.32	97.76	0.366
chb05	123,441	68.02	79.41	89.04	90.93	0.374
chb08	128,429	54.51	85.49	91.92	93.46	0.590
chb09	112,281	76.35	87.81	95.90	96.76	0.241
chb10	121,490	73.53	81.99	89.51	90.99	0.337
chb17	125,957	62.16	83.53	90.00	91.05	0.403
chb18	123,719	72.67	88.54	96.30	97.24	0.292
chb19	124,386	76.56	83.66	93.67	95.17	0.223
chb21	123,744	76.11	85.85	90.84	92.03	0.342
chb22	124,865	60.16	69.04	83.54	86.06	0.351
chb23	126,442	76.81	87.47	95.12	96.13	0.343
Avg.	–	68.97	84.75	92.63	93.86	0.354

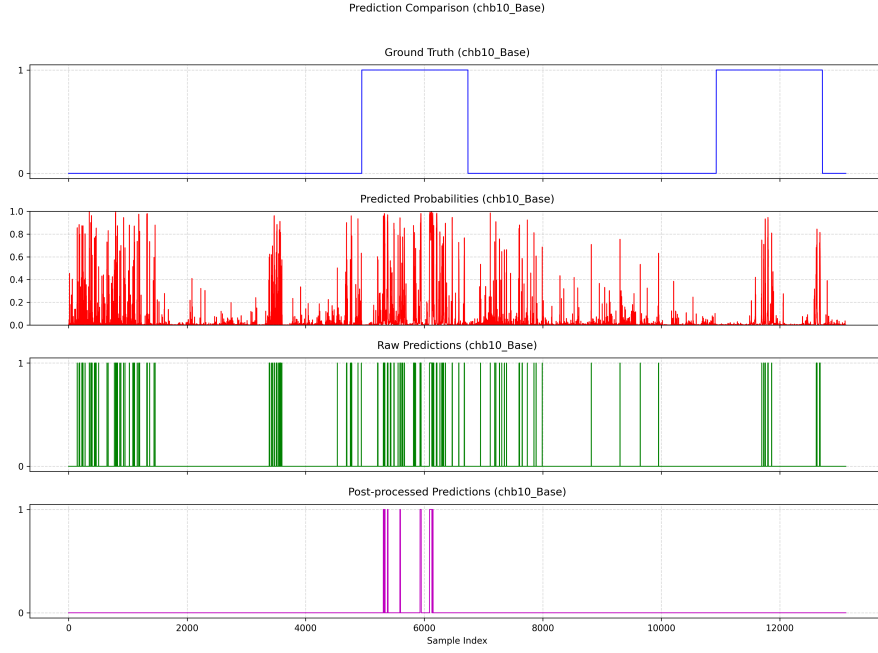
Observations:

- The LOPO model without any fine-tuning achieved an average accuracy of **68.97%**. Importantly, **10 out of 12 subjects** achieved LOPO accuracies above 60%, indicating moderate generalization to unseen individuals.
- Fine-tuning the model with just **100 patient-specific samples** increased the average accuracy significantly to **84.75%**, demonstrating the value of minimal personalization.
- With **1000 samples**, the average accuracy rose to **92.63%**, and further increased to **93.86%** when **2000 samples** were used.
- This shows that our CNN model, although trained on general population data, adapts remarkably well to unseen individuals with limited fine-tuning.
- Transfer learning proves to be a practical and effective method for bridging the gap between generalized and personalized seizure prediction systems.

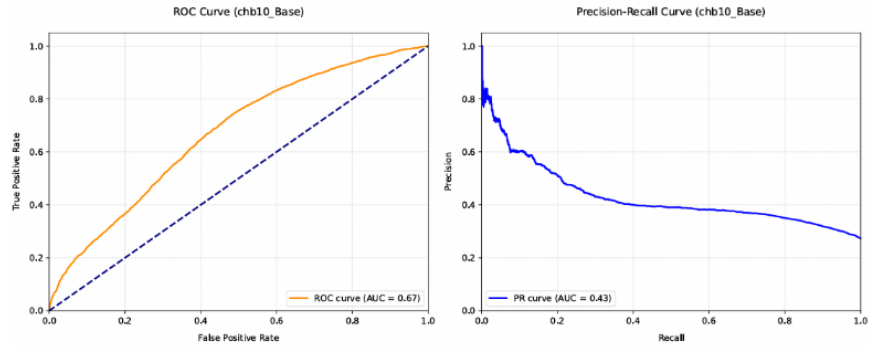
4.4.1 Visual Analysis for Subject chb10

To better understand the impact of transfer learning, we provide visualizations of the training performance for subject **chb10** at various stages: before and after applying transfer learning with 100, 1000, and 2000 samples.

4.4. Patient-Specific Evaluation using Transfer Learning

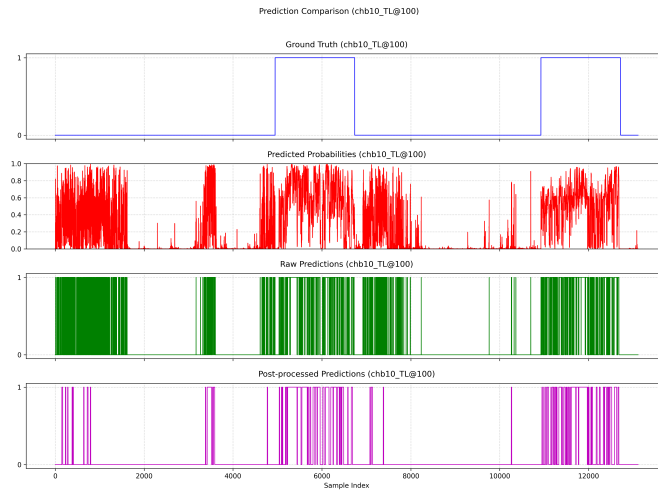


(a) Training and validation performance



(b) ROC and Precision-Recall Curves

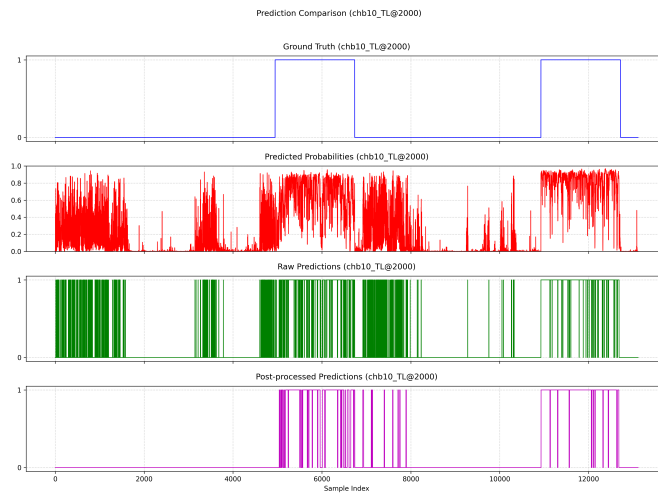
Figure 4.6: Model performance for chb10 before Transfer Learning (LOPO approach)



(a) Transfer Learning with 100 samples



(b) Transfer Learning with 1000 samples



(c) Transfer Learning with 2000 samples

Figure 4.7: Model performance for chb10 after Transfer Learning with different sample sizes

Table 4.6: Performance comparison of seizure prediction methods (PI = Patient-Independent, PS = Patient-Specific)

Study	Method	Approach	Acc. (%)	Sen. (%)	Spe. (%)	AUC
Dissanayake et al. [11]	Multitask CNN	PI	88.81	93.45	81.64	0.9273
	Siamese CNN	PI	91.54	92.45	89.94	0.9694
Usman et al. . [12]	Transfer Learning	PS	92.39	–	–	–
	CNN + SVM	PI	–	92.70	90.80	–
Pan et al.. [13]	CNN-4Res + CAD	PI	93.77	–	–	–
Narayana & Kodali . [14]	Hybrid LPF-LSTM	PI	98.40	97.80	96.20	0.9800
Zhang et al.. [8]	CNN + GAN	PS	92.20	–	–	–
Ours	PISNet	PI	98.54	98.94	98.41	0.9986
	PISNet-TL	PS	93.86	–	–	–

4.4.2 Discussion and Related Work

Table 4.6 provides a comprehensive performance comparison of recent seizure prediction methods with our proposed models, PISNet (patient-independent) and PISNet-TL (patient-specific via transfer learning).

Our patient-independent model, PISNet, outperforms all existing methods in terms of overall accuracy (98.54%), sensitivity (98.94%), specificity (98.41%), and AUC (0.9986). Compared to Dissanayake et al.’s multitask and Siamese CNN models, which reported 88.81% and 91.54% accuracy respectively, PISNet achieves a substantial improvement of over 6–9 percentage points in accuracy and offers significantly better balance between sensitivity and specificity.

The model also exceeds the performance of Narayana and Kodali’s LPF-LSTM approach, despite their model achieving 98.4% accuracy. PISNet maintains higher sensitivity and specificity while using a simpler 1D CNN architecture that processes raw EEG directly, avoiding complex handcrafted preprocessing.

Our patient-specific model, PISNet-TL, which applies transfer learning to the generalized model, achieves 93.86% accuracy even with limited subject data. This is comparable to or better than other patient-specific works such as Zhang et al.’s CNN+GAN method (92.20%) and Daoud and Bayoumi’s encoder-decoder + BiLSTM framework.

Trade-Offs and Interpretations:

While some patient-specific models in literature show high performance, they often require significant patient data and are not generalizable across subjects. In contrast, PISNet achieves state-of-the-art results in a patient-independent setting, making it more suitable for real-world applications where subject-specific EEG recordings may not be available beforehand.

A potential trade-off in generalized models is that initial performance for some subjects (e.g., chb01, chb08) may be lower in LOPO evaluation. However, with transfer

learning, these models quickly adapt — with accuracy rising from as low as 51.66% to over 98% for chb01, demonstrating excellent adaptability with just a few patient-specific samples.

Final Remarks and Future Outlook

Conclusion

This seminar project presented the design and evaluation of a compact and hardware-efficient seizure prediction model, termed **PISNet**, based on a 1D Convolutional Neural Network (1D-CNN). The core objective was to accurately distinguish between preictal and interictal EEG states while maintaining a lightweight structure suitable for deployment on digital hardware platforms.

Using the publicly available CHB-MIT Scalp EEG dataset, we developed a robust preprocessing pipeline involving downsampling, segmentation, and z-score normalization. Our proposed 1D-CNN model achieved **98.54% accuracy**, **98.94% sensitivity**, and an AUC of **0.9986**, outperforming several recent state-of-the-art seizure prediction models in both patient-independent and patient-specific setups.

The network was architected with RTL deployment in mind—filter sizes, layer depth, and pooling strategies were selected to balance predictive performance with resource constraints. These decisions position our model as a strong candidate for future FPGA-based medical AI accelerators.

Future Scope

This project lays the groundwork for developing real-time, low-power seizure detection systems using compact 1D-CNNs. Future work will focus on deploying the trained model at the RTL level, including model quantization for fixed-point representation, hardware implementation using Verilog or VHDL, and FPGA-based prototyping to evaluate performance metrics such as latency and resource usage. Additionally, the system will be extended to support multiple subjects while maintaining efficiency, and on-chip preprocessing modules will be integrated to handle raw EEG signals directly, thereby advancing toward a complete embedded hardware solution.

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